sheet-kit

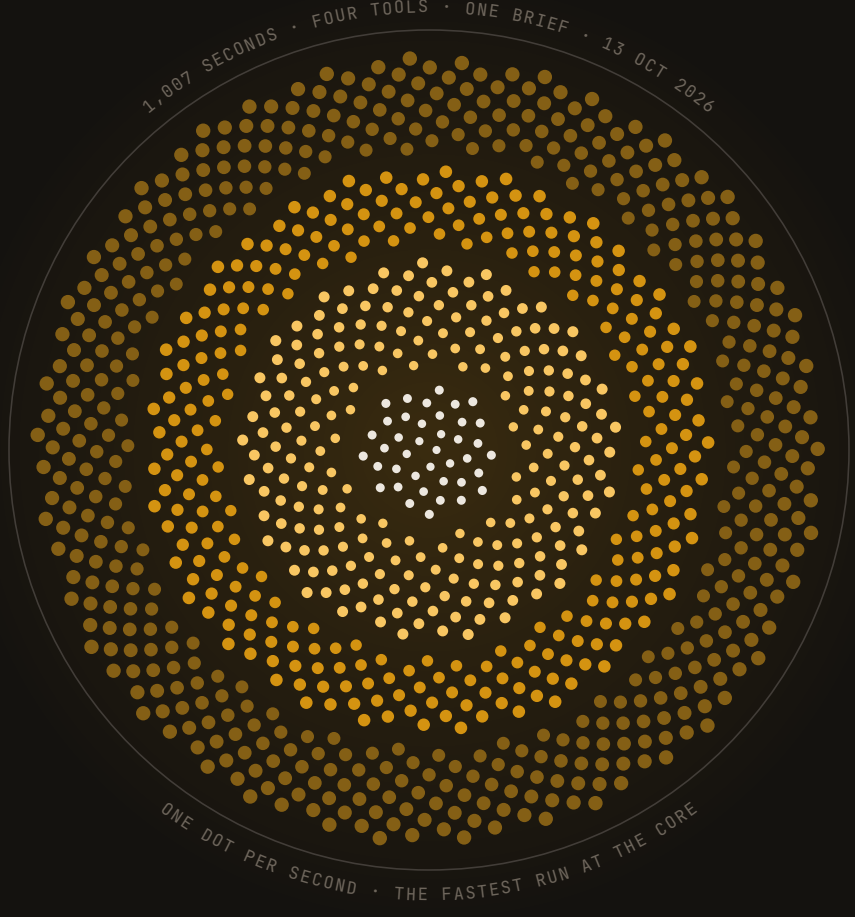
infographics, typeset.

A concept book drawn from one real run: four spec-driven tools, one brief, 13 October 2026. Every mark on these pages is set by code from the data.

CONTENTS

page · what is drawn

02 Pick a tool . . . . . decision ladder
03 The run, drawn . . . . . glyphs · orbit · units
04 Every command, one map . . . . . metro map
05 Where the gate lives, as flows . . . . . alluvial
06 The ecosystem . . . . . packed bubbles · timeline
07 The gate, engineered . . . . . state machine · code
08 Shapes of a run . . . . . radar · small multiples
09 Where approval lives . . . . . editorial long-read
10 Colophon . . . . . type · palette · QR



SUPERPOWERS	BMAD	OPENSPEC	SPEC KIT
40 s · $0.33	217 s · $1.89	277 s · $2.16	473 s · $2.84

FIG. 1 — 1,007 dots, one per second of the run, on a golden-angle spiral. The fastest run is the core; each slower one rings it.

ENGINE Typst 0.15.1 · cetz 0.5.3 · fletcher 0.5.8
TYPE Newsreader · Inter · JetBrains Mono
DATA The 13 Oct 2026 run · checked 21 Oct 2026

Match the process to the cost of being wrong.

Seven ways to run spec-driven development: which to pick, what to type, where the approval lives.

Which tool fits you

START HERE · IF NO, GO DOWN

IF YES

1 Small, solo, low stakes, and you keep the docs current?
→ **Plain AGENTS.md** — Install nothing

2 Must a human approve before any code? Money, data, health.
→ **sdd-kit** — Claude Code or Pi. Else p. 6

3 Should the editor run the whole flow, nothing to assemble?
→ **Kiro** — The editor runs the flow

4 Does your team think in roles: product, architecture, QA?
→ **BMAD** — Asks before it guesses

5 Does a team need one documented pipeline per feature?
→ **Spec Kit** — Budget the review time

6 What goes wrong today?

- Forgets decisions → **OpenSpec**
- Skips tests, steps → **Superpowers**
- Both, plus your gate → **OpenSpec + Superpowers**

Where the approval lives

TOOL	GATE LIVES IN	SKIPPABLE?
AGENTS.md plain memory	Nowhere	No gate to skip
OpenSpec spec-first, fluid	The prompt	Yes
Spec Kit ordered pipeline	The prompt (files checked)	Yes
Superpowers harness-first	The prompt (HARD-GATE)	Yes. It never ran in our test
BMAD role-first	The prompt (approval menu)	Yes. One line of pre-approval
Kiro IDE-native	A click in the IDE	Quick Spec skips it. Not tested
sdd-kit gated kit	A .status file on disk	Only if it ignores its skill

In all four tools we ran, the approval lived in the prompt. If the gate matters, put it on disk.

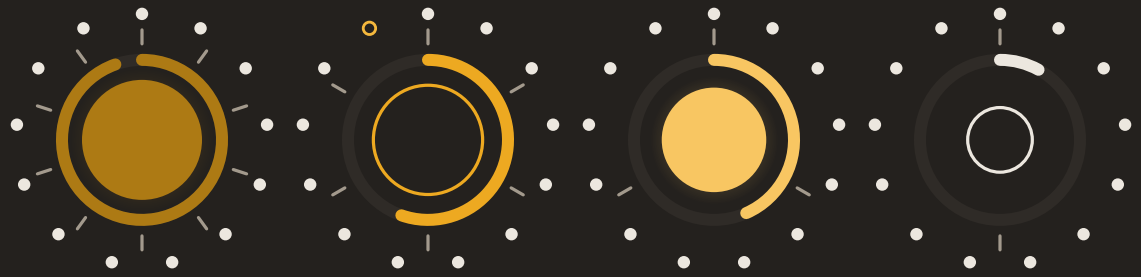
HOW, IN 14 LINES: PAGE 6

Same brief, four tools: the run, drawn.

8.6× apart on cost · 11.8× on time · 4 of 4 kept the gate in the prompt

FOUR TOOLS, ONE BRIEF

Each glyph is one run.



Spec Kit	OpenSpec	BMAD	Superpowers
$2.84 · 473 s	$2.16 · 277 s	$1.89 · 217 s	$0.33 · 40 s
10 prompts · 13/13	6 prompts · 12/13	3 prompts · 13/13	2 prompts · 13/13

HOW TO READ A GLYPH



1 CORE
Area is the cost. Filled: asked before coding.

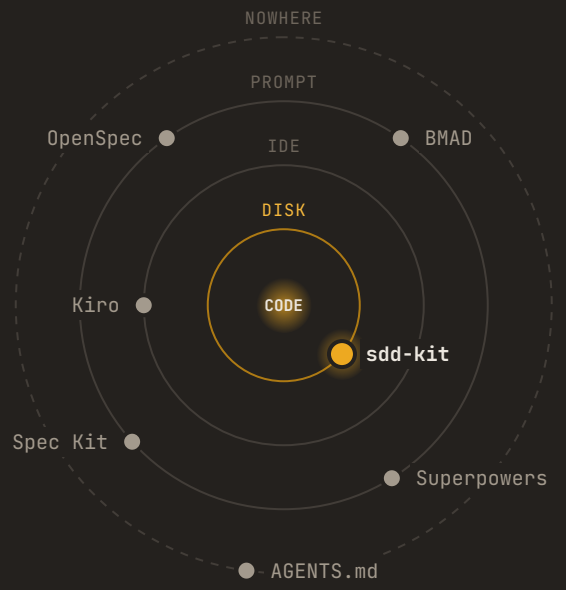
2 ARC
Time to a passing build. A full turn is 500 s.

3 RAYS
One ray per prompt you had to send.

4 RING
The 13 acceptance checks. Hollow is a miss.

WHERE THE GATE LIVES

How close it sits to the code.



The farther out, the easier it is to skip. Only one ring is a file the agent can't argue with.

1,007 SECONDS TO A PASSING BUILD

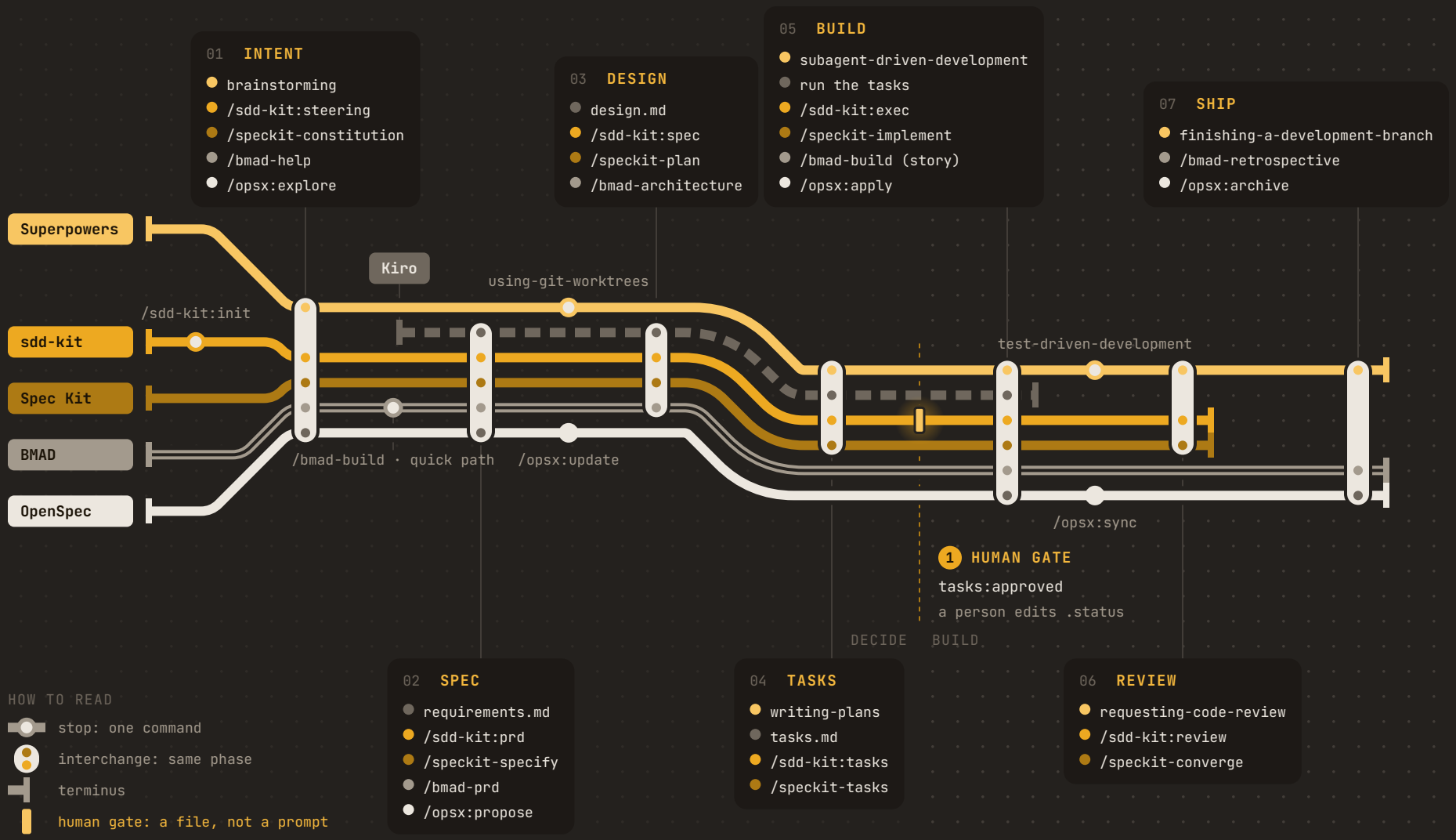
one dot = one second



COMMAND ATLAS

Every command, one map.

6 lines · 36 stops · 7 interchanges · 1 human gate



Superpowers	Kiro	sdd-kit	Spec Kit	BMAD	OpenSpec
7 stops · prompt	4 stops · IDE	7 stops · disk	6 stops · prompt	6 stops · prompt	6 stops · prompt

FLOWS

Where the gate lives, as flows.

7 tools · 4 places a gate can live · 1 on disk



WHERE IN THE WORKFLOW IT ASKS

◇ asked in the prompt □ a click in the IDE ◆ a token on disk

AGENTS.md	memory only: AGENTS.md, conventions.md, decisions.md	no gate
OpenSpec	explore — propose — update ◇ apply — sync — archive	skippable
Spec Kit	constitution — specify — plan — tasks ◇ implement — converge	skippable
Superpowers	brainstorming ◇ worktrees — writing-plans — subagents — TDD — code review — finish branch	skippable
BMAD	help — build ◇ prd — architecture — build (story) — retrospective	skippable
Kiro	requirements □ design □ tasks □ run	skippable
sdd-kit	init — steering — prd ◆ spec ◆ tasks ◆ exec — review	hard to skip

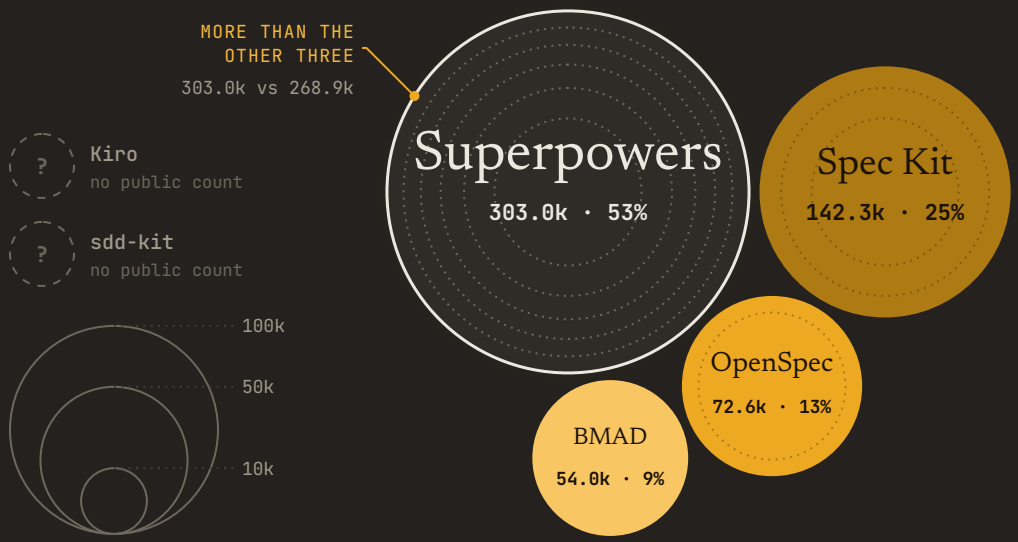
572k stars, four repos, four releases in ten days.

GitHub stars and releases · 21 Oct 2026

PACKED BUBBLES

Where the attention went.

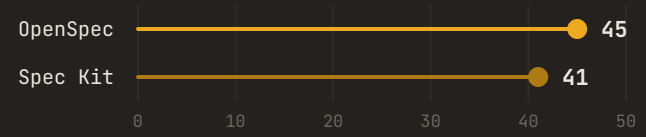
Each circle's area is a repo's GitHub stars.



Stars via gh api, 21 Oct 2026. Four repos, 571.9k in all.

LOLLIPOP

Agents each tool names.



Not counted: Superpowers, BMAD, Kiro, sdd-kit.

LEDGER

What moved since the run.

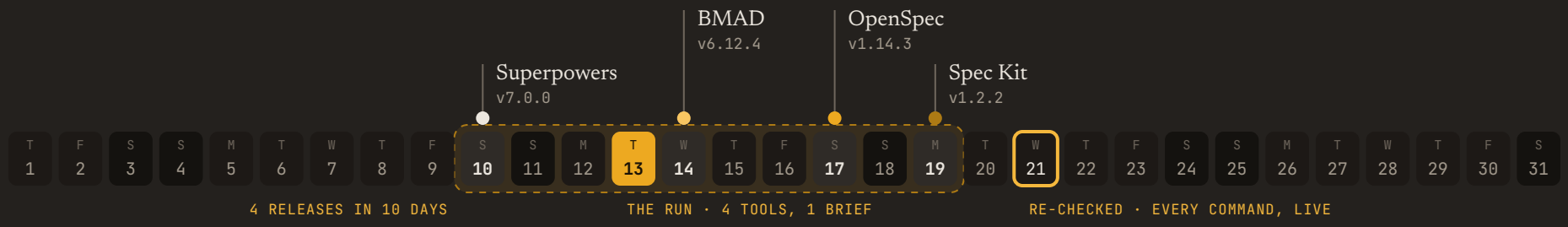
→ = new release

TOOL	AT THE RUN	ON 21 OCT	LICENCE
OpenSpec	1.14.1	→ 1.14.3	MIT
Spec Kit	1.2.0	→ 1.2.2	MIT
Superpowers	7.0.0	7.0.0	MIT
BMAD	6.12.2	→ 6.12.4	MIT
Kiro	not run	IDE 1.4.10	closed
sdd-kit	not run	0.1.0	MIT

CALENDAR STRIP · OCTOBER 2026

Three of four shipped after the run.

weekends dark · dot = release · amber = the run

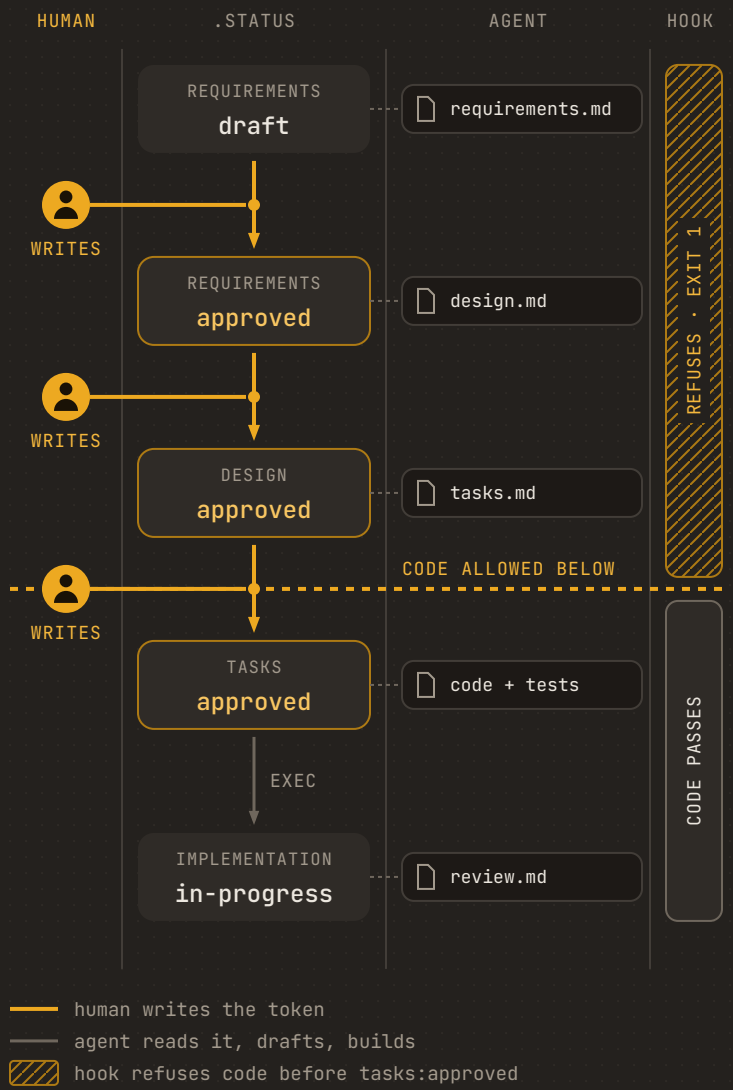


The gate, engineered.

Three pieces: a status file a human edits, four rules the agent reads, and a check that refuses to run.

STATE × ACTOR

Who moves the gate.



THE GATE AS CODE

```
01  # .ai/sdd/specs/001-feature/.status   (a human edits it)
02  requirements:draft                                          1
03  # then: requirements:approved -> design:approved
04  #       -> tasks:approved
05
06  # AGENTS.md (or CLAUDE.md)
07  ## Development flow
08  1. Every feature has a spec in `.ai/sdd/specs/NNN-name/`
09  2. Read the feature's `.status` BEFORE implementing anything  2
10  3. Never write code before `tasks:approved`                   3
11  4. Ambiguity during implementation: STOP and ask
12
13  # hook or CI step: block code until the token is there
14  grep -qxE 'tasks:approved|implementation:in-progress' \       4
15    "$SPEC/.status" || exit 1
```

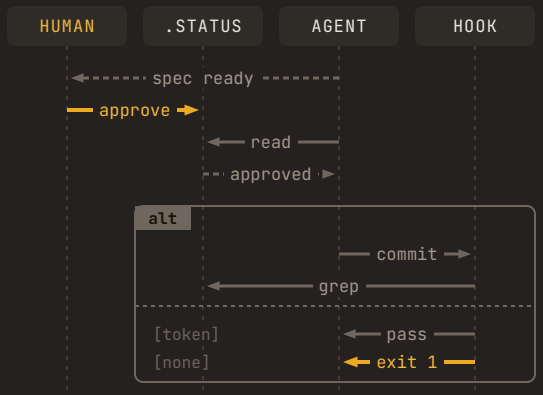
1 THE TOKEN
A human edits this line. The agent only reads it.

2 READ FIRST
The agent reads .status before any code.

3 THE RULE
No code before tasks:approved.

4 THE CHECK
A check that refuses to run: exit 1.

ONE APPROVAL, IN ORDER



ON DISK

```
your-repo/
├─ AGENTS.md                          4 RULES
└─ .ai/
   ├─ steering/                       4 FILES
   └─ sdd/specs/
      └─ 001-feature/
         ├─ .status                   THE GATE
         ├─ requirements.md
         ├─ design.md
         ├─ tasks.md
         └─ review.md
```

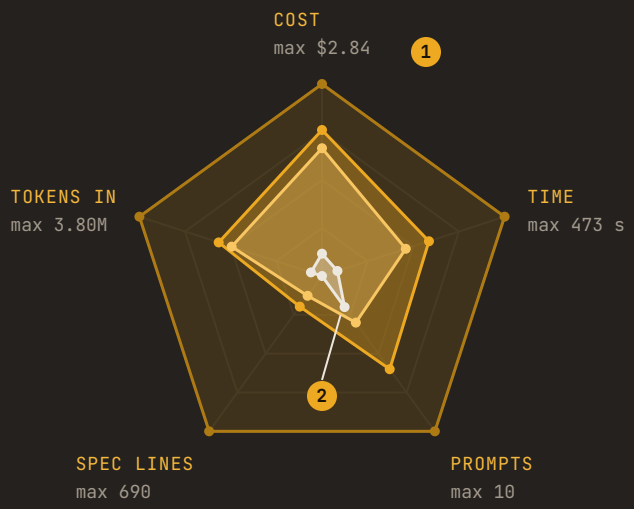
Shapes of a run: same brief, four footprints.

The 13 Oct run · n = 1 per tool

RADAR · OVERLAY

Four footprints on one scale.

Spec Kit · OpenSpec · BMAD · Superpowers



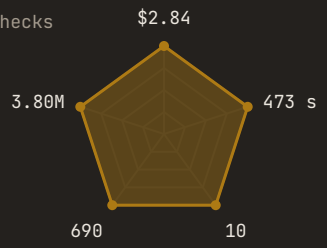
1. Spec Kit is the largest on all five axes: its shape is the whole pentagon.
2. Superpowers never passes 20% of the scale, on any axis.

RADAR · SMALL MULTIPLES

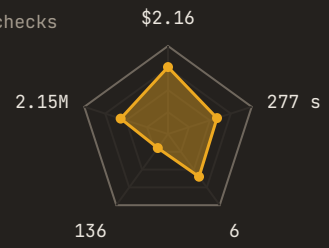
Each tool alone, real values at the tips.

same axes, same scale

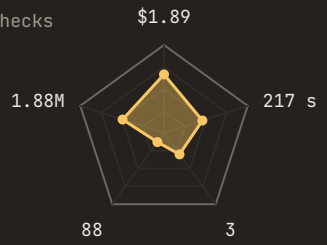
Spec Kit — 13/13 checks



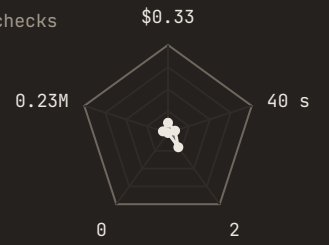
OpenSpec — 12/13 checks



BMAD — 13/13 checks



Superpowers — 13/13 checks

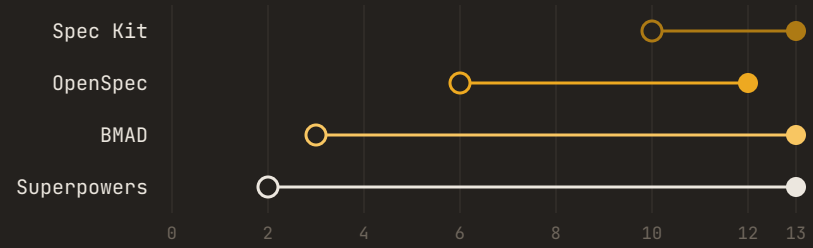


Scale: each value ÷ the largest of the four. All five maxima are Spec Kit's.

DUMBBELL

Prompts in, checks out.

○ prompts sent · ● checks passed



Checks are the 13-item acceptance list. More prompts bought no more checks.

QUADRANT

Asking first did not make it right.

the key: required or optional?

	DID NOT ASK	ASKED FIRST
RIGHT	**Superpowers** — *"Decided required, said so after the code"*	**BMAD** — *"Recommended required (right)"*
WRONG	**OpenSpec** — *"Guessed optional, flagged it before code"*	**Spec Kit** — *"Recommended optional (wrong)"*

FIELD NOTES · SPEC-DRIVEN DEVELOPMENT

Where *approval* lives.

Four spec-driven tools ran the same brief. The code converged, the bills ran 8.6 times apart, and in every run the step that mattered most sat where the agent could skip it.

By Felipe Fontoura · AI Engineer

4/4 tools kept the gate in the prompt

8.6× apart on cost for the same feature

Run 13 Oct 2026 · checked 21 Oct 2026

THE RUN

13 Oct 2026. claude-opus-5-5 in Claude Code 2.1.296, headless, a fresh clone per tool, scored on the same 13-item check.

THE BRIEF

An in-memory ledger: a 24-line `src/ledger.js`*, two tests, one base commit. Add an idempotent* `transfer()` *with a key.*

WHERE THE GATE LIVES

NOWHERE	AGENTS.md
PROMPT	OpenSpec, Spec Kit, Superpowers, BMAD
IDE	Kiro
DISK	sdd-kit

SECONDS TO A PASS

Spec Kit	473
OpenSpec	277
BMAD	217
Superpowers	40

NOTES

1 API-equivalent cost. n = 1.
2 Status values: the book, 3.5.
3 The hook is ours, not the book's.

IN ALL FOUR TOOLS WE RAN, the approval lived in the prompt. That sentence is the whole finding, and it took a ledger to get there: a twenty-four line file, two tests, one base commit, and a request for an idempotent transfer with a key. Two questions were left open on purpose: is the key required, and does a failed transfer use it up?

Each tool got a fresh clone, the same model and Claude Code 2.1.296, run headless. Spec Kit spent the most: `$2.84` over 473 seconds and ten prompts, with 690 lines of spec. OpenSpec came in at `$2.16`, BMAD at `$1.89` and Superpowers at `$0.33`, in 40 seconds and two prompts.[1] The code converged anyway. Three tools passed all 13 acceptance checks; OpenSpec passed 12.

WHO ASKED FIRST

Only BMAD asked both open questions before writing code, and it recommended the right answer: the key is required. Spec Kit asked before the plan, then recommended the wrong one. OpenSpec asked nothing and guessed, but flagged it before code. Superpowers decided the key was required on its own and said so after the code, because its brainstorming step never loaded.

The agent can misread a rule. It can't argue with an exit code.

A PROMPT CAN BE ARGUED WITH

Every one of those tools keeps its gate in text the agent reads. OpenSpec stops after a proposal and waits for your OK. Spec Kit refuses to implement until `plan.md` and `tasks.md` exist, but a one-line plan passes. Superpowers puts a HARD-GATE in its brainstorming skill. BMAD offers an approval menu the agent marks itself, and one line of pre-approval skips it. These are instructions, and an instruction is something a model weighs. Kiro moves the gate into its editor: a human click between phases, which no check can read.

A FILE CANNOT

A file on disk is different. Give each feature a `.status` file that only a human edits, with one token per phase.[2] Tell the agent, in AGENTS.md, to read it before implementing anything and never to write code before `tasks:approved`. Then add what a prompt cannot give you: a hook or CI step that greps the file and exits 1 when the token is missing.[3]

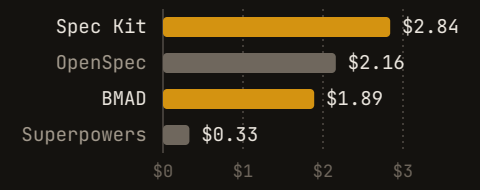


FIG. 1 *API-equivalent cost of one brief, 13 Oct 2026. Amber: the tool asked before coding.*

THE FIVE TOKENS, IN ORDER

1	`requirements:draft`	DRAFT
2	`requirements:approved`	YOU
3	`design:approved`	YOU
4	`tasks:approved`	YOU
5	`implementation:in-progress`	EXEC

The hook passes on 4 and 5 only.

WHAT THE TOTALS HIDE

Two details never reach a total. Spec Kit's analyze step caught four of its own tests that would pass before the feature existed. Superpowers, run by hand, asked first.

None of this needs a new tool. sdd-kit ships the `.status` file and the skills that read it; the hook is yours. Pick the lightest process that closes your gap, then decide where the yes is written down. ■

HOW THIS BOOK WAS MADE

Set in three voices, drawn by code.

Each page of this book is one Typst file: data in, PDF out, in about a second. No browser, no layout tool, and no mark placed by hand.

SCAN OR CLICK

The free SDD tools cheatsheet.

felipefontoura.com/get/sdd-tools-cheatsheet

Newsreader

HEADINGS · OPSZ 16

Regular	*Italic*	400
Medium	*Italic*	500
SemiBold	*Italic*	600

Aa Gg Qq & fi fl
0123456789

Production Type · OFL

Inter

BODY AND UI · OPSZ 14

Regular	*Italic*	400
Medium	*Italic*	500
SemiBold	*Italic*	600
Bold		700

→ ← ↑ ↓ ↗ ● ○ ✓
≈ ≠ ≤ ≥ ∞ π ★

Rasmus Andersson · OFL

JetBrains Mono

LABELS AND DATA

Regular	*Italic*	400
Medium		500
Bold		700

0O 1lI {} []
-> => != // ~$

JetBrains · OFL · calt off

Index

PAGE · WHAT IT DREW

01 Golden-angle spiral
02 Decision ladder
03 Glyphs · orbit · units
04 Metro map
05 Alluvial flows
06 Bubbles · timeline
07 State machine · code
08 Radar multiples
09 Long-read, drop cap
10 QR · specimen · palette

PALETTE · THE INK TOKENS

bg	#14120F	elevated	#1D1916	card	#24211E	soft	#2F2B27
border	#423D38	border-strong	#6F675D	paper	#ECE7DF	secondary	#A39A8D
muted	#6E665C	inverse	#1A1206	amber-300	#F8C662	amber-400	#F5B83D
amber-500	#EDA921	amber-600	#D49311	amber-700	#AD7A14		